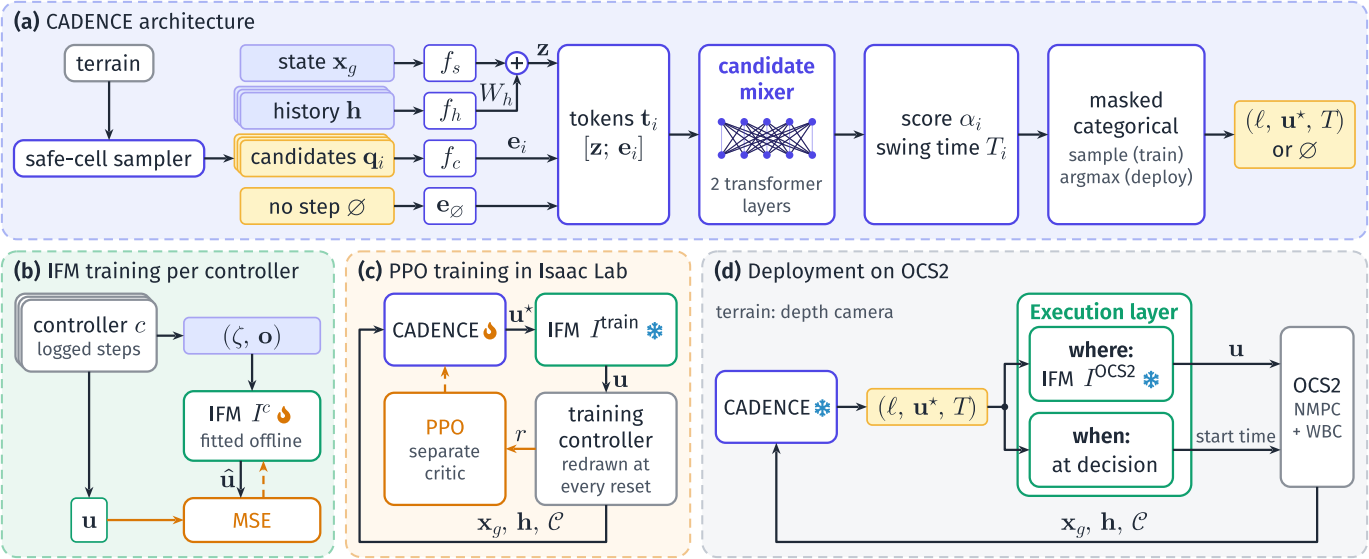

(a) CADENCE architecture
terrain
safe-cell sampler
state $\mathbf{x}_g$
history $\mathbf{h}$
candidates $\mathbf{q}_i$
no step $\varnothing$
$f_s$
$f_h$
$f_c$
$\mathbf{e}_\varnothing$
$W_h$
$\mathbf{e}_i$
$\mathbf{z}$
tokens $\mathbf{t}_i$
$[\mathbf{z};\ \mathbf{e}_i]$
candidate mixer
2 transformer layers
score $\alpha_i$
swing time $T_i$
masked categorical
sample (train)
argmax (deploy)
$(\ell,\ \mathbf{u}^\star,\ T)$ or $\varnothing$
(b) IFM training per controller
controller $c$
logged steps
$(\zeta,\ \mathbf{o})$
IFM $I^c$
fitted offline
$\hat{\mathbf{u}}$
$\mathbf{u}$
MSE
(c) PPO training in Isaac Lab
CADENCE
$\mathbf{u}^\star$
IFM $I^{\text{train}}$
$\mathbf{u}$
PPO
separate critic
$r$
training controller
redrawn at every reset
$\mathbf{x}_g,\ \mathbf{h},\ \mathcal{C}$
(d) Deployment on OCS2
terrain: depth camera
CADENCE
$(\ell,\ \mathbf{u}^\star,\ T)$
Execution layer
where:
IFM $I^{\text{OCS2}}$
when:
at decision
$\mathbf{u}$
start time
OCS2
NMPC
+ WBC
$\mathbf{x}_g,\ \mathbf{h},\ \mathcal{C}$